Shaikh Ahmed Azad

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Ankara, Turkey | *Civil Engineering Student — Computational Structural & Earthquake Engineering*

Education

Middle East Technical University (METU) **Ankara, Turkey**
B.Sc. Civil Engineering (4th Year) *Oct 2022 – Expected Sep 2026*

Technical Skills

Structural & Computational Analysis: Finite Element Analysis (FEA), Structural Dynamics (SDOF/MDOF), Seismic Design, Matrix Structural Analysis.
Programming & Software Development: Python (PyQt6, SciPy, ObsPy), MATLAB, Object-Oriented Design (OOP), GUI Development.
Engineering Software & BIM: SAP2000, ETABS, AutoCAD, Revit, RocScience, Navisworks, Synchro 4D, OpenRoads, HEC-RAS, MS Project, LaTeX.

Publications & Conference Submissions

Azad, S. A., Saldamlı, K. A., & Türer, A. (Submitted).
"Failure Mode and Capacity of Nailed/Screwed Timber CLT Slab Segments Under Four-Point Bending."
Undergraduate Researcher | Extended Abstract

- Submitted to **INTER 2026** (Norway); executed **displacement-controlled four-point bending tests** on mechanically fastened, recycled timber CLT segments to evaluate load-deflection responses.
- Analyzed composite action and stiffness using the **γ-method** (**TABY 2024**), achieving a **93% agreement** between analytical predictions and experimental results.
- Compared 4-nail, 5-nail, and 4-screw configurations, identifying that ultimate capacity is governed by **bending rupture** of outer laminations rather than fastener shear.

Honors & Competitions

DASK Earthquake Resistant Building Design | *National Finalist (ODTÜ Kuleleri)* **Feb 2026 – Present**

- Selected from a competitive national pool to construct and dynamically test a 30-story twin-tower model.
- Designed a coupled shear wall and macro-braced system with bridge links for torsional stability, sizing members via **TBDY 2018** Equivalent Seismic Load and modal analysis.

Computational Engineering Projects

OpenCivil: Structural Analysis Software | *Sole Developer* | *Project Website* **2024 – Present**

- **Engine & Architecture:** Open-source "glass box" 3D FEA desktop application built on MVC; ships **5 solvers** (Linear Static, Modal, RSA, LTHA, Solid FEM) with **6–7 decimal place agreement** to SAP2000 across complex section geometries including AISC profiles, trapezoidal, pipe, and tube.
- **FEM Formulations:** Implements **Timoshenko beam theory**, static condensation for member releases, cardinal insertion points, rigid end offsets, and **Tet10 quadratic solid elements**; global system assembled via **SciPy Sparse** solvers.
- **Seismic & Dynamics: LTHA** solver (Modal Superposition, Newmark-β) with real-time 3D earthquake response animation and **3D RSA** (CQC/SRSS), both **TBDY 2018** compliant; rendered through a custom **PyQt6 / OpenGL** pipeline.
- **Parametric API & Hybrid Interface:** Developed a `pip`-installable Python API (`pip install opencivil`) for headless batch analysis. Engineered an embedded CLI terminal within the GUI featuring real-time, two-way state synchronization, allowing users to seamlessly transition between visual modeling and programmatic execution.

CE-Star: Undergraduate Research Program | *Seismic Hazard & Signal Processing Software* **2025 – 2026**

- **Data Acquisition & Hazard:** Desktop platform with an interactive map interface for automated **AFAD** batch-downloading filtered via a custom M_w-distance matrix; integrates a full **PSHA hazard model** for site-specific seismic hazard visualization without external API dependency.
- **Signal Processing & IM Engine:** Pipeline for instrument response removal and computation of **5+ intensity measures** (Arias, CAV, Fourier, Housner, SED); **Newmark-β** integration for SDOF/MDOF elastic response spectra (S_a, S_v, S_d).
- **Reporting & SHM Application:** 1-click **TBDY 2018** compliant geospatial PDF report generator; applied to process strong motion datasets for a real-time monitored **216m building in Izmir** for rapid post-event condition assessment.

Professional Experience

DSI – State Hydraulic Works **Ankara, Turkey**
Civil Engineering Intern (Dams Department) *Aug 2025 – Sep 2025*

- Validated hydraulic design calculations and reviewed technical design files for large-scale dam construction projects.
- Conducted on-site inspections to correlate technical design specifications with field implementation realities and construction methodologies.

Parabol Yazılım (METU Technopolis) **Ankara, Turkey**
Urban Mobility Intern *Jul 2025 – Aug 2025*

- Analyzed large-scale pedestrian and micromobility datasets to assess infrastructure performance against Level of Service (LOS) benchmarks.
- Submitted technical reports proposing evidence-based infrastructure improvements to enhance urban safety and circulation efficiency.